

Rolando Medina Rosner

Full-stack Software Engineer

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Summary

Full-stack engineer with 3 years building and modernizing the hospital information system used across Paraguay's national public hospital network. On a three-person team, I led its migration off legacy Java/jQuery to React and TypeScript, introduced its containerized deployment and CI/CD, and handle most of its frontend, backend internals, and deployment work.

Experience

Software Engineer, Taiwan ICDF — National Hospital Information System – Asunción, Paraguay Mar 2024 – present

- Own inpatient clinical modules end-to-end (physician orders, diagnoses, nursing workflows, pharmacy and medication requests, medical-history PDF export), used daily by clinical staff across the national public hospital network.
- Rebuilt the largest legacy module from unmaintainable JSP and jQuery into structured TypeScript and restructured its Struts backend into a DAO/DTO JSON API; replaced webpack with rsbuild to cut build times.
- Evaluated Lit and React for the frontend rewrite, proposed React after Lit could not meet SSR requirements, built the entire project setup, and migrated the three most complex modules (physician, nursing, pharmacy) to production with only minor launch-day issues.
- Building an offline-first PWA so rural hospitals keep operating through multi-day connectivity loss, using an IndexedDB outbox, service-worker background sync, and version-tagged DTOs that migrate stale client data on reconnect.
- Replaced script-based WAR deployments with a two-image Docker Compose setup; prototyped a k3s cluster and evaluated OpenYurt for edge-deployed hospital servers.
- Self-hosted an LLM and wrote the schema, retrieval, and prompt layer enabling natural-language-to-SQL queries over the clinical database.

DevOps Engineer Jr. (previously Frontend Developer Intern), ROSHKA S.A. – Asunción, Paraguay Aug 2023 – Feb 2024

- Built Jenkins CI/CD pipelines using Git Flow and GitHub Flow, containerized services with Docker, and integrated SonarQube quality gates.
- Contributed React frontend work to a banking web application as an intern; hired into a full-time engineering role at its conclusion.

Projects

Text-to-SQL RAG

Self-hostable natural-language-to-SQL assistant — [code](#)

- Answers plain-language questions by writing safe, read-only SQL, with defense-in-depth safety (SELECT-only DB user, sqlglot AST validation, enforced LIMIT), 4-tier RAG retrieval, SSE streaming, and an MCP server.
- FastAPI, Python, React 19, ChromaDB, Ollama/vLLM; runs fully self-hosted with no external API.

D y R Transportes

Production logistics platform — [code](#)

- In daily production use by a grain-transport company for trip, cargo, and driver-payment records, including the reports required by the national transport regulator.
- React, TypeScript, Vite, and Material UI (X Data Grid/Charts) frontend; Flask and MySQL backend; self-hosted with multi-arch Docker and Jenkins pipelines, built solo.

Sentiment Analysis on Financial News

June 2024

Final degree project

- Collected an original financial-news dataset and compared machine-learning methods (finBERT, pyABSA) for classifying article tone against stock-price movement; flagged by the department as one of the standout projects of the 2024 class.

Education

Universidad Politécnica Taiwán – Paraguay (UPTP), BSc in Computer Engineering

2020 – 2024

- GPA 4.81 / 5.00. Exchange student at National Taiwan University of Science and Technology (NTUST), 2022–2023.

Skills

Frontend: TypeScript, React, Next.js, PWA and service workers, IndexedDB, Tailwind CSS, Material UI, shadcn/ui

Backend: Java (Struts, Spring Boot), Python (Flask, FastAPI), REST API design, SQL, MySQL, PostgreSQL, C/C++

Infrastructure: Docker, Kubernetes (k3s), Terraform, GitLab CI, Jenkins, Proxmox, Linux

AI / Data: PyTorch, scikit-learn, pandas, NumPy, transformer models, self-hosted LLMs (vLLM), RAG

Languages: Spanish (native), English (TOEFL iBT 109/120), German (B1), Mandarin (elementary)

Certifications: NVIDIA DLI Fundamentals of Deep Learning; Machine Learning (Stanford); Statistics 110 (Harvard)